



< Previous



Next >

A formal definition of unexpectedness

Bookmark this page

Question: Unexpectedness should be a statistical notion, shouldn't it?

Answer: No! Statistics may even play no role whatsoever. For instance, many unexpected events are unique, unheard-of.

Unexpectedness is the core ingredient of relevance and subjective information.

Unexpected situations are situations that require complex circumstances to occur, while being simple to describe in comparison.

We already defined unexpectedness as the difference between **expected complexity** and **description complexity**. Now, we revisit this definition by observing that expected complexity, for rational agents, takes the form of causal complexity C_w . We may write:

Unexpectedness

$$U = C_w - C.$$

Unexpectedness

0/1 point (ungraded)

What is the unexpectedness of the New Jersey Lottery event? (see preceding exercise) (we ignore the complexity of locating the event in time and space)

☐

0 bits.

☐

0 or 10 bits, depending on the event we consider.

☐

10 bits.

☒

10 or 20 bits, depending on the event we consider.

☐

20 bits.

✖

Answer

Incorrect:
10 bits is the correct answer.
Note that if we consider the event "741 twice", causal complexity is about 20 bits, as we saw. Description complexity, on the other hand, is about 10 bits, as it must include the definition of 741.
If we consider the event "same number twice", causal complexity is about 10 bits, while description complexity is about 0 bits (no constant involved, just a copy operation).

Submit

You have used 1 of 1 attempt

Blasé attitude

1/1 point (ungraded)

Can you describe the "Blasé" attitude (most events aren't worth noticing) in complexity terms? The "Blasé" attitude amounts to ...

- ☐ 1a. increasing the causal complexity of all events
- ☐ 1b. diminishing the causal complexity of all events
- ☐ 2a. increasing the description complexity of all events
- ☐ 2b. diminishing the description complexity of all events
- ☐ 1a + 2a
- ☐ 1a + 2b

☒ 1b + 2a

☐ 1b + 2b



Answer

Correct:

The main point of the "blasé" attitude is to decrease causal complexity. One strategy for this is called 'trivialization'. It consists of situating the event within a broader class of events so as to increase its frequency of occurrence (e.g. "It's something that happens in many couples.") If the event is frequent, the value of its expected (or causal) causality must not be that large.

Submit

You have used 1 of 1 attempt

✓ Correct (1/1 point)

Paranoia and surprise

0/1 point (ungraded)

When adopting a paranoiac attitude, you are not surprised by events such as accidents that affect you. Why is it so?

☐ 1a. because the paranoiac attitude augments the causal complexity of such events

☐ 1b. because the paranoiac attitude diminishes the causal complexity of such events



☐ 2a. because the paranoiac attitude augments the observed complexity of such events

☐ 2b. because the paranoiac attitude diminishes the description complexity of such events

☐ 1a + 2a

☐ 1a + 2b

☒ 1b + 2a

☐ 1b + 2b



Answer

Incorrect:

From a paranoid perspective, chance does not exist. Events that affect you have low causal complexity, as they have been generated intentionally. Moreover, their target is simple by construction, as it is necessarily you.

A similar description can be made about invocations of fate ('It was meant to be').

Submit

You have used 1 of 1 attempt

 Answers are displayed within the problem

Fortuitous encounter

1/1 point (ungraded)

Coming across a celebrity by chance near your doorstep might be perceived as an interesting event. The story would be even better if you live in the suburbs rather than downtown. Why?

☒ It increases the causal complexity of the event.

☐ It diminishes the causal complexity of the event.

☐ It increases the description complexity of the event.

☐ It diminishes the description complexity of the event.



Answer

Correct:

The description complexity of the location remains simple in both cases, since it is your home. However, the causal complexity of bringing the celebrity close to your home is larger if you live in the suburbs (note that this causal complexity is roughly equivalent to the description complexity of your address from her perspective, as explained [there](#)).

Submit

You have used 1 of 1 attempt

✔ Correct (1/1 point)



[Open edX](#)
[Careers](#)
[News](#)

Legal

[Terms of Service & Honor Code](#)
[Privacy Policy](#)
[Accessibility Policy](#)
[Trademark Policy](#)
[Sitemap](#)
[Cookie Policy](#)
[Your Privacy Choices](#)

Connect

[Idea Hub](#)
[Contact Us](#)
[Help Center](#)
[Security](#)
[Media Kit](#)

